



Type ZWC De-energized Tap-changer

Product Structure

- 1) The tap-changer is of a split structure and consists of two parts: the body and the drive mechanism. The transformer manufacturer is responsible for connecting these two.
- 2) The tap-changer is of linear type and equipped with two insulating support plates as a frame. It is used in power transformers and large-scale distribution transformers. It is installed in the space between the windings and the tank wall. It benefits from a compact size and reduced tapping lead.
- 3) The tap changer operation is controlled by a Geneva wheel to ensure accurate positioning of the moving and fixed contacts.
- 4) The appropriate connection between the tap changer contacts is done through a classic clamping-type floating point method to meet the the current carrying capacity.
- 5) When necessary the tap-changer can be equipped with a shielding structure. This provides a uniform electric field is uniform and good partial discharge performance.
- 6) The positioning pin avoids the change of tap positions during installation, ensuring installation correctness and product reliability.
- 7) The manual drive mechanism is designed for side operation. Moving from one position to the next requires one revolution. The drive handle is fixed to its end cap and can be used as a position locking component. It also has provision for an external lock.
- 8) The drive shaft is suitable for installation on both drum and bell-type transformers. It has a telescopic drive shaft with a telescopic distance of 200mm allowing for simple installation.



Drive Mechanism



Tap-changer body

Technical Data

1	Tap-changer Model			ZWCI 500	ZWCI 1000
2	Phases			Single phase	
3	Rated frequency（Hz）			50/60	
4	Rated through current（A）			500	1000
5	Short-circuit current test（kA）	Thermal（3s）		8	15
		Dynamic（Peak）		12.5	25
6	Insulation level	Voltage（kV）		138	138
		The highest voltage for equipment（kV）		145	145
		Rated separate source AC withstand voltage（1min）（kV）	To ground or phase	275	275
			Between contact	50 with 50mm spacing	50 with 50mm spacing
				80 with 80mm spacing	80 with 80mm spacing
			Between max. & min. taps of the same	100	100
			Rated lightning impulse withstand voltage（1.2/50μs）（kV）	To ground or phase	650
		Between contact		140 with 50mm spacing	140 with 50mm spacing
				225 with 80mm spacing	225 with 80mm spacing
		Between max. & min. taps of the same phase		250	250
7	Drying temperature（℃）			Vacuum drying 110℃，Vapor phase drying 125℃	
8	Connection mode			Linear	
9	Operating method			Side manual operation	
10	Max. operating positions			5	
11	Contact resistance between fixed and moving			≤350μΩ	
12	Partial discharge（pC）			≤50	
13	Leakage Test			No leakage under 6x10 ⁴ PA for 24h	
14	Mechanical life			More than 10,000 operations	
15	Temperature rise of contact at 120% of rated			Below 15K（in oil）	

Note: Users should consider the insulation distances of tap-changer to the ground, to coils and other accessories. Enough space must be offered during design.

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